Distributed Algorithms of Finding the Unique Minimum Distance Dominating Set in Directed Split-Stars

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Abstract

A distance-k dominating set S of a directed graph D is a set of vertices such that for every vertex v of D, there is a vertex $u \in S$ at distance at most k from it. Minimum distance-k dominating set is especially important in communication networks for distributed data structure and for server placement. In this paper, we shall present simple distributed algorithms for finding the unique minimum distance-k dominating set for $k = 1, 2$ in a directed split-star, which has recently been developed as a new model of the interconnection network for parallel and distributed computing systems.

Keyword: Distributed algorithms, Distance-k dominating set, Interconnection networks, Directed split-stars.

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1 Introduction

Star graphs provide attractive interconnection scheme for parallel computers and distributed networks, hence characterizations of this architecture has been widely investigated [1, 2, 6, 10, 11]. In [5], Cheng et al. gave a variant architecture of star graphs which is namely split-stars. Later on, Cheng and Lipman [4] proposed an assignment of orientation to the split-stars as a new class of digraph called the directed split-stars. They also showed that the resulting digraphs are not only strongly connected, but, in fact, they have maximal arc-fault tolerance and a small diameter. Recently, Wang et al. [13] showed that the directed split-stars have the unique minimum distance-k dominating set for $k = 1, 2$. In this paper, we shall present distributed algorithms for finding such sets.

Let $D = (V, A)$ be a directed graph (digraph) with vertex set V and arc set A, where $A \subseteq V \times V$. An arc $<u, v>$ is said to be *directed* from u to v, in which case we say that u *dominates* v, and v is *dominated* by u. Let k be a positive integer. A set S of vertices in a digraph D is called a *distance-k dominating set* if for every vertex v of D, there is a vertex $u \in S$ at distance at most k from it. A *minimum distance-k dominating set* is a distance-k dominating set with the minimum cardinality.

Researches on domination and related topics have much been attracted by graph theorists for their strongly practical applications and theoretical interesting. Efficient construction for distance-k dominating sets can be applied in the context of distributed data structure [12], where it is proposed that a distance-k dominating set can be selected for locating copies of a distributed directory. Likewise, such a set is useful for efficient selection of network centers for server placement, where it is desired to ensure that each node in the network is sufficiently close to some server (cf. [3]). A thorough treatment of domination in graphs and digraphs can be found in [7, 8].

The remaining part of this paper is organized as follows. In Section 2, we give the definition of directed split-stars and introduce some basic terminologies and notations. Besides, we also define a model of the system under investigation. Section 3 presents the distributed algorithms for finding the unique minimum distance-k dominating set for $k = 1, 2$, and shows the correctness of the proposed algorithms. Finally, a concluding remark is given in the last section.

2 Preliminaries

For a digraph $D = (V, A)$ and $u, v \in V$, the *distance* $d(u, v)$ is the number of arcs along a shortest path from u to v, and $d(u, v) = \infty$ if there is no path from u to v in D. Let k be a positive integer. The *distance-k outset* of a vertex u in D is the set $O_k(u) = \{v \in V \mid d(u, v) = k\}$, while the *distance-$k$ inset* is $I_k(u) = \{v \in V \mid d(v, u) = k\}$. For simplicity, we write $O(u)$ and $I(u)$ for $k = 1$. Let $od(u)$ and $id(u)$ denote the *outdegree* and the *indegree*, respectively, of a vertex $u \in V$, i.e., $od(u) = |O(u)|$ and $id(u) = |I(u)|$. For terms of digraphs not defined here please refer to [9].

The *n-dimensional directed split-star* $\overrightarrow{S_n^2}$ is a digraph whose vertices are in a one-to-one correspondence with $n!$ permutations $[p_1, p_2, \ldots, p_n]$ of the set $N = \{1, 2, \ldots, n\}$, and two vertices u, v of $\overrightarrow{S_n^2}$ are connected by an arc $<u, v>$ if and only if the permutation of v can be obtained from u by either a 2-exchange or a 3-rotation. Let $u = [p_1, p_2, \ldots, p_n]$. A 2-*exchange* interchanges the first symbol p_1 with the second symbol p_2 whenever $p_1 > p_2$, i.e., $v = [p_2, p_1, \ldots, p_n]$. A 3-*rotation* counterclockwise rotates the symbols in positions 1, 2 and i for some $i \in \{3, 4, \ldots, n\}$, i.e., $v = [p_i, p_1, p_3, \ldots, p_{i-1}, p_2, p_{i+1}, \ldots, p_n]$. Figure 1 depicts an example of $\overrightarrow{S_n^2}$ for $n = 4$.

Let $u = [p_1, p_2, \ldots, p_n]$. It is easy to see from the definition that u has the outdegree equal to $n-1$ and the indegree equal to $n-2$ if $p_1 > p_2$; for otherwise, u has the outdegree

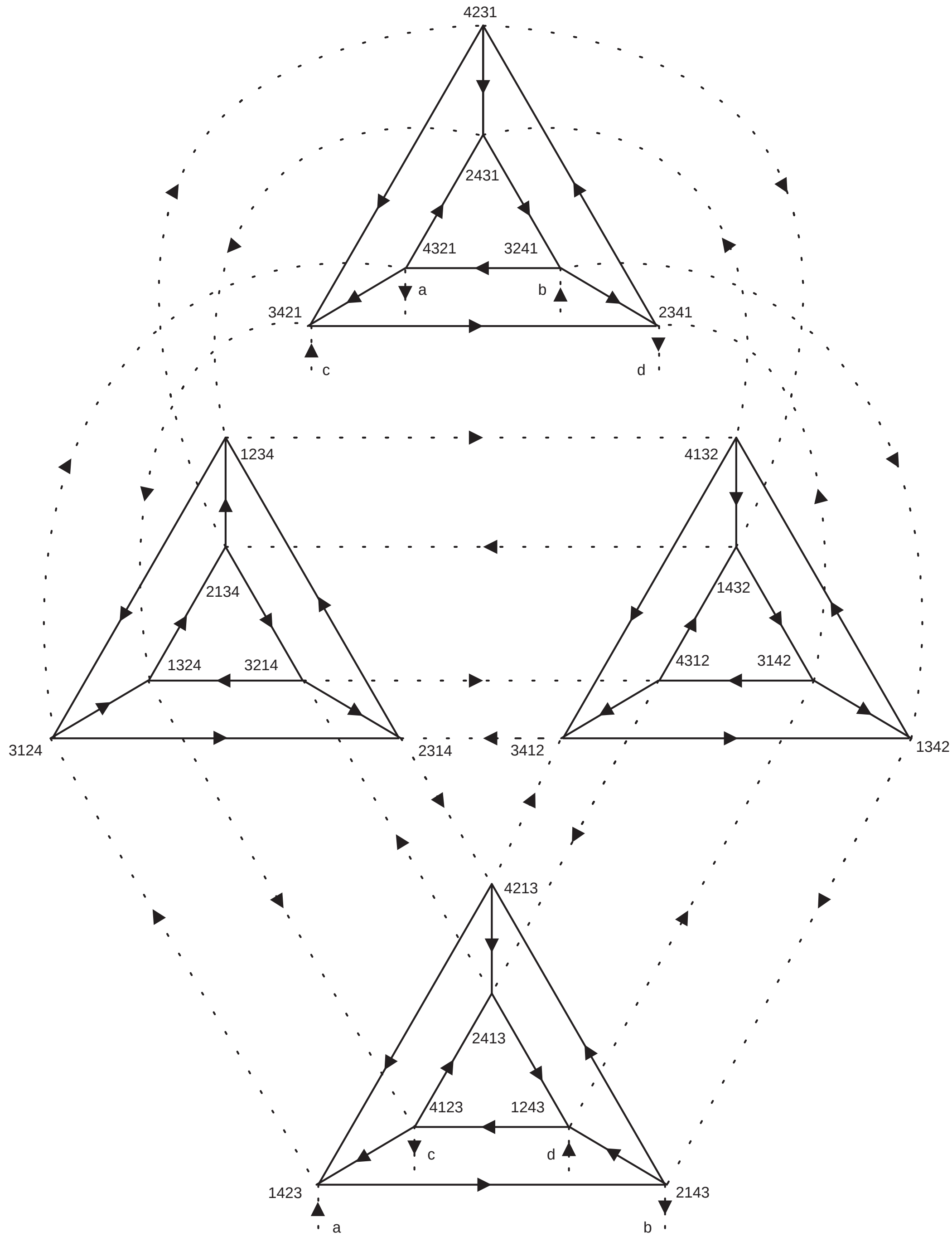

Figure 1: 4-dimensional directed split-star.

equal to $n-2$ and the indegree equal to $n-1$. To simplify the description of our results, we define the following sets. Let $N_i = N \backslash \{i\}$ for each $i = 1, 2$, and let $N_{12} = N \backslash \{1, 2\}$, where $N = \{1, 2. \ldots, n\}$. The following result presented in [13].

Theorem 1 *Let* $S = \{[p_1, 1, p_3, \ldots, p_n] \,|\, p_i \in N_1, i \in N_2\}$ *and* $T = \{[2, 1, p_3, \ldots, p_n] \,|\, p_3, \ldots, p_n \in N_{12}\}$. *Then,* S *and* T *are the unique minimum distance-k dominating set of* $\overrightarrow{S_n^2}$ *for* $k = 1, 2$, *respectively.*

A distributed network can be viewed as a system consisting two types of components: processors and interconnections between processors. Usually, such a system is modeled as a digraph for which each vertex representing a processor, and a directed arc between processors representing message channel for communications. We define a *distributed algorithm* to be a collection of local algorithms from the same copy that one for each processor. Every processor execute its local algorithm and cooperates with the other processors to achieve the objective of the distributed algorithm. To communicate among themselves, the processors exchange information through messages. That is, the processors can communicate only by sending and receiving messages over the communication links. We assume that communications performed by the network is asynchronous, i.e., the transmitting message is placed in a queue waiting for the receiver to accept it and the sending processor can proceed immediately. We also assume that each vertex (processor) u knows the graph dimension n, the indegree $id(u)$, and the outdegree $od(u)$, while the identities of all vertices in the graph are unknown. Note that each processor has a local state and all variables are locally maintained by each process.

3 The distributed algorithms

For each processor x, we define the following variables referred by the algorithm. $x.indegree$ and $x.outdegree$ are two constants denoting the indegree and the outdegree of x, respectively. The boolean variable $x.member_k$ indicates whether x is a member of the unique minimum distance-k dominating set. Also, we have the following three variables: $x.data$ contains the message that can be transferred between x and its neighbors; $x.count$ records the number of incoming messages; and $x.sum$ is an auxiliary variable that can be used to accumulate information sent from the neighbors directed to x.

The algorithm of computing the distance-1 dominating set can be represented by the following two stages and it is independently executed on each processor. In the first stage, each processor set up the message and sends to its neighbors. In the second stage, the received messages are compared with the indegree for determining the membership of dominating set.

Algorithm DISTANCE-1 DOMINATION

begin

stage 1. $x.sum := 0$; $x.count := 0$; $x.member_1 :=$ FALSE;

if $x.outdegree < x.indegree$ then $x.data := 1$

else $x.data := 0$;

send $x.data$ to each vertex $y \in O(x)$;

stage 2. repeat

if there is an incoming message sent from the vertex $y \in I(x)$ then

$x.sum := x.sum + y.data$;

$x.count := x.count + 1$;

until $x.count = x.indegree$;

if $x.sum = x.indegree$ then $x.member_1 :=$ TRUE;

end.

Lemma 2 *Let* $S = \{[p_1, 1, p_3, \ldots, p_n] \,|\, p_i \in N_1, i \in N_2\}$. *Then* $x \in S$ *if and only if* $od(y) < id(y)$ *for every* $y \in I(x)$.

Proof. Suppose $x = [p_1, 1, p_3, \ldots, p_n]$ and $y \in I(x)$. It is clear that x is a 3-rotation neighbor of y. Let $y = [1, p_i, p_3, \ldots, p_{i-1}, p_1, p_{i+1}, \ldots, p_n]$ for some $i \in N_{12}$. Since the symbol 1 is on the first position of y, $od(y) = n-2 < id(y) = n-1$.

Conversely, we assume $x \notin S$ and we will show that there is a vertex $y \in I(x)$ such that $id(y) < od(y)$. Consider the following two cases.

Case 1: the symbol 1 is on the first position of x. Let $x = [1, p_2, p_3, \ldots, p_n]$ and $y = [p_2, 1, p_3, \ldots, p_n]$. Clearly, x is a 2-exchange neighbor of y. Since the symbol 1 is on the second position of y, $id(y) = n-2 < od(y) = n-1$.

Case 2: the symbol 1 is on the ith position of x where $3 \leq i \leq n$. Let $x = [p_1, p_2, p_3, \ldots, p_{i-1}, 1, p_{i+1}, \ldots, p_n]$ and $y = [p_2, 1, p_3, \ldots, p_{i-1}, p_1, p_{i+1}, \ldots, p_n]$. Clearly, x is a 3-rotation neighbor of y. Since the symbol 1 is on the second position of y, $id(y) < od(y)$.

Q. E. D.

Theorem 3 *The Algorithm* DISTANCE-1 DOMINATION *correctly determines the unique minimum distance-*1 *dominating set of* $\overrightarrow{S_n^2}$.

Proof. For each processor x, we have $x.outdegree < x.indegree$ if x is a 2-exchange neighbor of some vertex and $x.indegree < x.outdegree$ otherwise. From the algorithm, it is easy to see that the variable $x.sum$ indicates the number of vertices directed to x with the outdegree less than the indegree. By Lemma 2, x is a member of the unique minimum distance-1 dominating set if and only if each vertex directed to x has the outdegree less than the indegree, and therefore it can be determined by the last statement of stage 2.

Q. E. D.

We now give the complexity analysis as follows. Since the message transferred once only occurs in the first stage of the algorithm, the communication complexity is evidently proportional to the number of directed arcs in the digraph.

For computing the distance-2 dominating set of $\overrightarrow{S_n^2}$, we design the algorithm by three stages where the first two stages are the same with the previous one, excepting that the accumulated information is sent out to the neighbors of x again (see the last statement of stage 2). Then, the third stage compares the secondly received messages with certain condition for determining the membership of distance-2 dominating set. Since the algorithm receives message twice on each processor and these incoming messages must be discriminated for controlling the sequence of stages, we may assume that every message provides with a tag for indicating the different type of messages. For convenience, we omit the tag filed in the outgoing messages (because the sequence of outgoing messages can be handled by the algorithm), but use $y.data_1$ and $y.data_2$ as the description of tag field in the incoming messages. In addition, the accumulated information of x in stage 3 is recorded in the variable $x.total$.

Algorithm Distance-2 Domination

begin

stage 1. $x.sum := 0$; $x.count := 0$; $x.member_1 :=$ FALSE;

if $x.outdegree < x.indegree$ then $x.data := 1$

else $x.data := 0$;

send $x.data$ to each vertex $y \in O(x)$;

stage 2. repeat

if there is an incoming message sent from the vertex $y \in I(x)$ then

$x.sum := x.sum + y.data_1$;

$x.count := x.count + 1$;

until $x.count = x.indegree$;

if $x.sum = x.indegree$ then $x.member_1 :=$ TRUE;

send $x.sum$ to each vertex $y \in O(x)$;

stage 3. $x.total := 0$; $x.count := 0$;

repeat

if there is an incoming message sent from the vertex $y \in I(x)$ then

$x.total := x.total + y.data_2$;

$x.count := x.count + 1$;

until $x.count = x.indegree$;

if $x.total = \frac{(n-2)(n-3)}{2}$ and $x.member_1 =$ TRUE then

$x.member_2 :=$ TRUE;

end.

Lemma 4 *Let* $S = \{[p_1, 1, p_3, \ldots, p_n] \,|\, p_i \in N_1, i \in N_2\}$ *and* $x \in S$. *The following statements are true:*

(i) *If* $y \in I_2(x)$ *and* $y = [y_1, y_2, \ldots, y_n]$, *then* $y_1 \notin \{1, p_1\}$.

(ii) *If* $Y = \{y \in I_2(x) \,|\, od(y) < id(y)\}$, *then* $|Y| = [1 + 2 + \cdots + (n-2)] - (n - p_1)$.

Proof. Let $x = [p_1, p_2, p_3, \ldots, p_n]$. Since $p_1 > p_2 = 1$, x cannot be a 2-exchange neighbor of any vertex. Suppose $y \in I_2(x)$ and let w be a vertex in $I(x)$ such that $y \in I(w)$. Since x is a 3-rotation neighbor of w, we have

$$w = [p_2 = 1, p_i, p_3, \ldots, p_{i-1}, p_1, p_{i+1}, \ldots, p_n] \quad \text{for some } i \in \{3, 4, \ldots, n\}. \tag{1}$$

Regardless of the fact that w is either a 2-exchange or a 3-rotation neighbor of y, it is easy to see that p_i must be put in the first position of y. Since $p_i \in N_1$ and $p_i \neq p_1$ for $i = 3, \ldots, n$, this shows that statement (i) holds.

To show the statement (ii), we first define $Y_w = \{y \in I(w) \,|\, od(y) < id(y)\}$ for each vertex $w \in I(x)$. We now consider a vertex $y \in Y_w$. Since $od(y) < id(y)$, it implies that y is a 2-exchange neighbor of some vertex. Thus, if y_1 and y_2 are the symbols on the first position and the second position of y, then $y_1 < y_2$. In particular, w is a 3-rotation neighbor of y in this case. According to (1), we can represent

$$y = [p_i, p_j, p_3, \ldots, p_{j-1}, p_2 = 1, p_{j+1}, \ldots, p_n] \quad \text{where } j \in N_2 \backslash \{i\}. \tag{2}$$

By the fact $p_i < p_j$, we have $|Y_w| = n - p_i$. Indeed, the above argument shows that all vertices in Y_w have the same symbol, say p_i, on the first position. Thus $Y_w \cap Y_{w'} = \emptyset$ for any two vertices $w, w' \in I(x)$. Note that

$$\begin{aligned} Y &= \{y \in I_2(x) \mid od(y) < id(y)\} \\ &= \bigcup_{w \in I(x)} \{y \in I(w) \mid od(y) < id(y)\} \\ &= \bigcup_{w \in I(x)} Y_w \end{aligned}$$

Therefore, we have the following consequence.

$$|Y| = \sum_{w \in I(x)} |Y_w| = \sum_{p_i \in N_1 \backslash \{p_1\}} n - p_i = \Big[\sum_{k=2}^{n} (n-k)\Big] - (n - p_1)$$

This completes the proof.

Q. E. D.

In the following proof we assume that $S = \{[p_1, 1, p_3, \ldots, p_n] \mid p_i \in N_1, i \in N_2\}$ and $T = \{[2, 1, p_3, \ldots, p_n] \mid p_3, \ldots, p_n \in N_{12}\}$. Clearly, $T \subset S$ and the following corollary directly follows from statement (ii) of Lemma 4.

Corollary 5 *Suppose $x \in S$ and let $Y = \{y \in I_2(x) \mid od(y) < id(y)\}$. Then $x \in T$ if and only if $|Y| = 1 + 2 + \cdots + (n-3)$.*

Theorem 6 *The Algorithm* Distance-2 Domination *correctly determines the unique minimum distance-2 dominating set of $\overrightarrow{S_n^2}$.*

Proof. Since $T \subset S$, the vertices of T are filtered through S by the algorithm. Let x be a vertex in S. Since $x.total$ indicates the number of vertices with the distance 2 directed to x and satisfying the property that the outdegree is less than the indegree, the correctness of the algorithm directly follows from Corollary 5 and the last statement of stage 3.

Q. E. D.

4 Concluding Remark

Wang et al. [13] showed that a directed split-star has the unique minimum distance-k dominating set for $k = 1, 2$. In this paper, we propose distributed algorithms for finding such dominating sets. However, the problem of finding a minimum distance-k dominating set for $k \geq 3$ on directed split-stars is still unsolvable.

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